

PRE-CLASS READING 1 OF 5

Co-Intelligence: Living and Working with AI

Ethan Mollick (2024). *Portfolio* / Penguin Random House. ISBN 9780593716717.

ASSIGNED READING

Introduction — “Three Sleepless Nights”	pp. xi–xix
Chapter 1 — “Creating Alien Minds”	pp. 1–26
Chapter 3 — “Four Rules for Co-Intelligence”	pp. 46–63
Chapter 6 — “AI as a Coworker” (read pp. 130–137 closely; skim the rest)	pp. 123–158

≈ 87 pages · approx. 2 hours

Note on chapter selection. You are jumping 1 → 3 → 6. Chapters 2, 4 and 5 are excellent but cover ground this course handles elsewhere. Read them if you have time.

KEY TAKEAWAY

AI’s capability boundary is **jagged, not linear** — superhuman at some tasks, incompetent at adjacent ones that look easier, with no reliable way to predict which is which from the outside. Mollick’s response is empirical rather than theoretical: you map the frontier by testing it, and you organise your work around where it actually falls.

WHY IT MATTERS FOR THIS COURSE

This is the anchor reading for the whole module. Mollick’s central claim — that AI capability does not track human intuitions about which tasks are hard — is the reason Class 1 spends its time on a **taxonomy**. If difficulty predicted performance, you could simply ask “is this hard?” It doesn’t, so you instead ask “**what kind of system is this, and what is that kind structurally good and bad at?**” The taxonomy is a deliberate substitute for an unreliable gut.

A1 then closes the loop: it makes you test the boundary empirically — a scikit-learn classifier against a foundation-model API — rather than reason about it, because even with the taxonomy, the frontier’s exact position for *your* task is only knowable by experiment. **Class 4’s** agent build is Centaur/Cyborg design under a different name. **Class 5’s** economics traces back to the BCG study in Chapter 6.

MAIN POINTS

Introduction — the calibration problem

- **Three sleepless nights.** “*I believe the cost of getting to know AI — really getting to know AI — is at least three sleepless nights.*” Not rhetoric: casual use produces a systematically wrong impression of capability in both directions. Sustained, adversarial, task-realistic use is the only route to a usable mental model. Approach the in-class Colab this way — probe it, don’t just run it.
- **AI as a General Purpose Technology.** Placed alongside steam, electricity and the internet: value comes from diffusion across every sector, not one killer application. Historically these take decades to register in productivity statistics. Sets up Class 5.
- **The exam evidence, read skeptically.** AI has “blown through both the Turing Test and the Lovelace Test” and “aces our hardest exams, from the bar exam to the neurosurgery qualifying test.” Note the gap this opens: passing an exam built for humans measures something much narrower than competence. The course returns to this when we cover evals.

Chapter 1 — what the thing actually is

- **A history that argues a point.** 1770s Mechanical Turk (a chess “automaton” concealing a human operator) → Turing and the imitation game → symbolic AI → statistical machine learning → the Transformer. Every era overestimated the near term and underestimated the long term. A caution against both hype and dismissal.

- **How it works, non-technically (p. 8).** Training corpora, the Transformer, and the attention mechanism — how the model weights which parts of the input matter when predicting the next token. Your on-ramp; Karpathy (Reading 3) does it properly.
- **Where the data came from (p. 12).** Unresolved copyright and consent questions attached to training corpora. Foreshadows Class 6 governance.
- **Plausibility ≠ accuracy (p. 25).** “Demonstrations ... can seem more impressive than they actually are because they are so good at producing answers that sound correct, at providing the illusion of understanding.” Paired with: “Where AI works best, and where it fails, can be hard to know in advance.” The most important sentence in the assigned pages.
- **The alien mind (p. 26).** The metaphor the book is built on: something that “acts very much like a person, but in ways that aren’t quite human.” Not a mind, not a database, not a search engine. Class 1’s taxonomy is an attempt to be precise about what it is.

Chapter 3 — the operating manual

- **Rule 1: Always invite AI to the table (p. 47).** Default to trying AI on any task, because you cannot predict where it helps without testing.
- **The Jagged Frontier (pp. 47–48).** Two distinct causes: **architecture** (tokens, not words — so counting and character-level tasks break) and **empirical unpredictability** (skill doesn’t track human-perceived difficulty). The frontier also moves with every model release, so any map you draw is dated.
“The AI is great at the sonnet, but because of how it conceptualizes the world in tokens rather than words, it consistently produces poems of more or less than fifty words. Similarly, some unexpected tasks (like idea generation) are easy for AIs while other tasks that seem to be easy for machines to do (like basic math) are challenges for LLMs. To figure out the shape of the frontier, you will need to experiment.”
- **Rule 2: Be the human in the loop (p. 52).** Because the failure mode is confident wrongness, verification cannot be delegated to the system producing the output.
- **Rule 3: Treat AI like a person — but tell it what kind of person it is (p. 56).** Persona framing measurably changes output quality. Feeds directly into Class 3’s prompting work.
- **Rule 4: Assume this is the worst AI you will ever use (p. 62).** Design decisions should anticipate capability improving, not assume today’s ceiling. Directly relevant to build-vs-buy in Class 2 and Class 5.

Chapter 6 — how the work gets divided (read pp. 130–137 closely)

- **Task taxonomy.** Sort work into **Just Me** (human judgment, ethics, accountability), **Delegated** (AI drafts, human verifies) and **Automated** (AI runs unsupervised). The sort is task-by-task, not role-by-role.
- **Centaur (pp. 130–137).** Strategic division of labour with a clear seam — human does strategy, AI does execution, discrete handoffs. Named for the mythical figure’s clean human/horse boundary.
- **Cyborg (pp. 130–137).** “Cyborgs blend machine and person, integrating the two deeply.” Interleaved at subtask level — starting a sentence for the AI to finish, iterating a prompt in real time. No clean seam.
- **Which mode when.** Centaur suits well-bounded tasks where you can predict what AI handles well; Cyborg suits exploratory work where the boundary is unknown. The choice is downstream of the frontier: the better you’ve mapped it, the more Centaur becomes viable.
- **The BCG field study.** Dell’Acqua et al., *Navigating the Jagged Technological Frontier* — pre-registered randomised trial, **758 BCG consultants**, three arms (no AI / GPT-4 / GPT-4 + prompt training):

Inside the frontier 18 realistic consulting tasks	12.2% more tasks completed · 25.1% faster · >40% higher quality, vs. control.
Outside the frontier 1 task designed to fall outside	AI users 19 percentage points less likely to reach the correct answer — control 84.5% correct vs. 60% and 70% for the AI arms. AI didn’t merely fail to help; it actively degraded performance.

Levelling effect

Below-threshold performers improved **43%**; above-threshold **17%**. Both groups improved significantly — the gap narrowed, the top didn't stall.

READING FOCUS QUESTIONS — bring your answers to class

1. Name one task in your own work where you'd expect AI to be *inside* the frontier, and one where you'd expect it *outside*. What's your reasoning — and how would you actually test it?
2. Is your current AI use Centaur or Cyborg? Would the other mode work better?
3. The BCG outside-frontier result showed AI making people *worse*. What would that look like in your industry?

SOURCE LINKS

Publisher (canonical citation) · penguinrandomhouse.com · ISBN 9780593716717

Buy (Asia) · amzn.asia/d/0iLjZgZY

Free Introduction excerpt · [Penguin Random House Higher Education](#)

Author's Substack · [One Useful Thing](#)

CAVEATS

1. **This summary is not a substitute for the reading.** It is a map, not the territory. Read the assigned chapters — and keep reading the rest of the book as the course progresses. Several chapters skipped here become relevant in Classes 3, 5 and 6.
2. **This summary was generated and fact-checked using Anthropic models.** Page numbers, quotations and chapter attributions were verified against the published text and the publisher's excerpt. Errors are still possible — the book is the authority. Flag anything that looks wrong.